

## **$E$ is Irrational**

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If the number  $e$  is defined to be the sum of the series  $\sum_{n=0}^{\infty} \frac{1}{n!}$ , then the error in approximating  $e$  by the partial sum  $1 + 1 + 1/2 + \cdots + 1/n!$  is

$$\begin{aligned} 0 &< e - \left(1 + \frac{1}{1!} + \cdots + \frac{1}{n!}\right) \\ &= \frac{1}{(n+1)!} + \frac{1}{(n+2)!} + \cdots \\ &= \frac{1}{(n+1)!} \left(1 + \frac{1}{n+2} + \frac{1}{(n+2)(n+3)} + \cdots\right) \\ &\leq \frac{1}{(n+1)!} \sum_{k=0}^{\infty} \left(\frac{1}{n+2}\right)^k \\ &= \frac{n+2}{(n+1)!(n+1)} \end{aligned}$$

The series converges quickly. Also, if  $e = a/b$  for positive integers  $a$  and  $b$ , then the number

$$x = b! \left( e - \left( 1 + \frac{1}{1!} + \cdots + \frac{1}{b!} \right) \right)$$

is an integer, and

$$0 < x \leq \frac{b!(b+2)}{(b+1)!(b+1)} = \frac{b+2}{(b+1)^2} < 1$$

which is a contradiction.